

Samuel Schiavone

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PROFILE

I am a researcher, programmer, and mathematician who combines mathematical abstraction with computational tools in order to discover, analyze, and elucidate. I have experience in learning abstract results and applying them to produce practical solutions.

SKILLS

Programming Languages: Python, PostgreSQL, SageMath, Magma, R

Programming Tools: Git, Flask, Jinja, GNU Parallel, Scikit-learn, Pandas

Languages: English (native), French (fluent), Spanish (intermediate)

SELECTED PROJECTS

Development for the L-functions and modular forms database (LMFDB) May 2018 – Present
Full Stack Web Developer Remote

- Collaborated on a joint open-source project to create an accessible repository of data in algebra and number theory, used by over 30,000 researchers and educators in math
- Co-created two sections of an online database: added data to the backend using Python and PostgreSQL; designed webpages to display the data using Flask, Jinja, and HTML
- Supervised four undergraduate students in the creation of a Magma and SageMath package to compute additional data
- Connected to other sections of the LMFDB by identifying and linking to related objects using scripts written in Python and SageMath

Computations in Inverse Galois Theory Jan 2022 – Nov 2024
Academic Researcher Cambridge, MA

- Collaborated on a team of 6 researchers to solve an open mathematical problem in computational number theory
- Combined theoretical expertise with practical programming skills in Python, PostgreSQL, SageMath, and Magma to produce a research article and two repositories of code, one a general Magma package and one tailored to a specific example
- Presented results in two invited talks to audiences of 30-50 researchers at VaNTAGe and Simons AGNTC annual meeting
- Synthesized detailed understanding of theoretical results in arithmetic geometry with technical programming expertise to give an explicit solution to an open mathematical problem

EXPERIENCE

Research Scientist Sep 2019 – Aug 2025
Massachusetts Institute of Technology Cambridge, Massachusetts

- Conducted research in computational arithmetic geometry
- Devised and implemented algorithms to compute mathematical objects, wrote mathematical proofs, and chronicled my research in academic articles
- Aided in the development and maintenance of the L-functions and Modular Forms Database (<https://www.lmfdb.org/>), working both with data in PostgreSQL and Python on the backend and in Flask, Jinja, and HTML on the frontend
- Instructed undergraduate students in courses on linear algebra, algebraic geometry, mathematical communication, and Belyi maps

EDUCATION

Dartmouth College	Hanover, New Hampshire
<i>Ph.D. in Mathematics</i>	Sep 2019
The University of Vermont	Burlington, Vermont
<i>M.S. in Mathematics</i>	Dec 2013
Amherst College	Amherst, Massachusetts
<i>B.A. in Mathematics, magna cum laude</i>	May 2010